

HRMS — Product Requirements Document

Building Smarter Workplaces.

This PRD takes the vision, architecture, and feature list already worked out and turns it into a requirements document: who WorkForge is for, what it must do and in what order, how success will be measured, and what's deliberately left out of v1. The architecture document remains the technical source of truth for how the system is built; this document focuses on what and why. Additions that go beyond the original material — goals/metrics, personas, non-functional requirements, a permissions matrix, risks, and open questions — are marked as PM recommendations so it's clear what's carried over versus newly proposed.

Contents

1. Executive Summary
2. Problem Statement
3. Vision & Positioning
4. Goals & Success Metrics
5. Users & Personas
6. Scope & Roadmap at a Glance
7. Functional Requirements
8. Roles & Permissions
9. Non-Functional Requirements
10. System Overview & Technology Stack
11. Data Model Summary
12. Release Plan & Milestones
13. Risks, Assumptions & Open Questions
14. Appendix: Glossary & Reference Documents

1. Executive Summary

WorkForge is a workforce operating platform, not a records-only HRMS: it pairs core HR operations — employee records, attendance, leave, payroll — with automation and an AI-driven insight layer (predictive attrition, CV matching, sentiment analysis, promotion recommendations) on one role-aware platform. It targets two markets with the same core product: Nigerian public-sector organizations, starting with a possible pilot across Ministries, Departments and Agencies (MDAs), and private-sector organizations that need the same rigor at a different scale.

The product ships in three phases so a solid, trustworthy operational core (Phase 1) is live and producing real data before the AI/analytics layer (Phase 3) is built on top of it — not the other

way around. This document defines what each phase must deliver, who it's for, how it will be measured, and what's explicitly out of scope for v1.

2. Problem Statement

HR administration in most Nigerian public-sector and mid-size private organizations still runs on paper, spreadsheets, and disconnected tools:

Fragmented records. Employee data is scattered across departments/MDAs — even basic questions like "how many staff do we have, and where?" take days to answer accurately.

Unverifiable attendance. Sign-in sheets and self-reported hours are vulnerable to time fraud and give management no real visibility into who's actually at work on a given day.

Slow, manual leave processing. Paper-based approvals and manually tracked balances mean slow turnaround and frequent disputes.

Manual, error-prone payroll. Spreadsheet-based payroll is slow to run and hard to audit — a real risk given statutory deductions (tax, pension, NHF) that must be accurate.

No shared recruitment pipeline. Applicant data, interview notes, and hiring decisions live in individual inboxes rather than a trackable, auditable system.

No real-time leadership visibility. Workforce cost, attrition, and performance are reported late, manually, and inconsistently.

Commodity software. Most existing HR tools in this market store data but don't help anyone act on it — a digital filing cabinet, not a decision-support system.

WorkForge is designed to close all seven gaps on one platform, for both government MDAs and private employers.

3. Vision & Positioning

Vision: Building Smarter Workplaces. An intelligent HR management system that helps organizations manage employees, automate HR operations, analyze workforce performance, and make data-driven decisions.

Positioning: WorkForge should be positioned as an AI-powered Workforce Operating Platform built for African organizations** — not "another HR system." That framing is deliberate: it's the difference between a system that stores employee data and one that actively helps HR, managers, and executives make better decisions, and it's the story that should carry through an academic submission, employer conversations, and any government stakeholder pitch.

Reference use cases (both first-class, not one primary/one secondary):

Public sector: centralized MDA records, transparent recruitment, automated attendance, digital leave, audit trails, workforce analytics for planning and budgeting.

Private sector: faster onboarding, self-service HR, workforce dashboards, compliance-ready document management, scalability from startup to enterprise.

4. Goals & Success Metrics

Goal	Metric	Current baseline	Illustrative target
Faster leave decisions	Request → approval/decision turnaround	Days (paper-based)	Under 24 hours
Faster, cleaner payroll	Payroll cycle processing time	Multi-day manual reconciliation	Under 1 business day, zero manual recalculation
Trustworthy attendance	Disputed/corrected attendance records per month	Not currently measured	Under 2% of records
Lighter HR admin load	HR staff time on manual data entry/reporting	Not currently measured	–50%
Real self-service adoption	% of employees active on the self-service portal	0% (new system)	Over 80% within 3 months of rollout
Leadership visibility	Time to answer "what's our headcount / payroll cost / attrition right now?"	Manual report, days	Under 1 minute via Executive Dashboard
Platform reliability	Uptime	N/A	99%+ during pilot, 99.5%+ once payroll-critical

5. Users & Personas

Role	Who they are	What they need from WorkForge	Primary touchpoints
Employee	Any staff member, MDA or company	Self-service for leave, payslips, profile, quick answers	Employee Portal, AI Assistant
Manager	Team/unit lead	Visibility into their team's attendance, leave, performance; fast approvals	HR Dashboard (team-scoped)
HR Officer	Runs day-to-day HR operations	One system for records, recruitment, leave policy, documents	HR Dashboard, Recruitment, Documents
Payroll/Finance Officer	Runs payroll and financial reporting	Accurate, automated, auditable pay runs	Payroll module, Reports
Director / Executive	MDA head or company leadership	A real-time, high-level view of the workforce for decisions	Executive Dashboard
Applicant	External candidate	A simple way to apply and track application status	Public Recruitment Portal
Super Admin	Platform/IT owner	Tenant setup, global configuration, security oversight	Admin Dashboard

This table reconciles two slightly different role lists in the source material — "Employees, Managers, HR Officers, Directors, Payroll Officers, Applicants, Super Admin" and "Admin, HR, Manager, Employee, Finance, CEO" — into one canonical set. "Director" and "CEO" are treated as the same role type at different organization types; "Finance" and "Payroll Officer" are treated as the same role.

6. Scope & Roadmap at a Glance

6.1 In scope, by phase

| **Phase** | Theme | Modules | Priority |

|---|---|---|---|

| **Phase 1** — Core HR Platform | Get the operational basics right and trustworthy | Auth & Access Control · Employee Management · Department & Org Structure · Attendance (QR/GPS) · Leave Management · Payroll (basics) · Dashboards & Reporting | P0 — must-have |

| **Phase 2** — Enterprise Features | Round out the employee lifecycle | Recruitment (core) · Performance Management · Learning Portal · Document Management · Notifications · Asset Management · Digital Employee ID | P1 — should-have |

| **Phase 3** — Innovation Layer | Differentiate with AI/analytics, built on Phase 1–2 data | AI HR Assistant · Predictive Attrition · Smart Recruitment (CV matching) · Sentiment Analysis · Promotion Recommendation Engine · Smart Attendance enhancements · Executive Dashboard · Automated Workflows · Multi-Org Support (user-facing) | P1/P2 — flagship differentiators, some showcase-only |

6.2 Out of scope for v1

- Multi-country payroll/tax logic (v1 is Nigeria-only: PAYE, pension, NHF)
- Deep accounting/ERP/GL integration (v1 exports reports; direct integration is a later phase)
- Native iOS/Android apps (v1 is a responsive web app)
- Union/collective-bargaining-agreement management
- Full benefits marketplace/brokering (v1 handles benefits as payroll deductions, not enrollment shopping)
- Complex org-restructuring workflows (mergers, mass transfers) beyond basic department/role reassignment
- Dedicated biometric hardware integration (turnstiles, standalone scanners) — v1 uses device cameras/QR only

7. Functional Requirements

Each module lists its objective, key capabilities, and priority. P0 = must-have for the phase to be considered done, P1= should-have, P2 = stretch/showcase.

7.1 Phase 1 — Core HR Platform

- Authentication & Access Control — P0

Secure, role-based sign-in for every user type, with session management.

- Login via email/username + password; JWT access + refresh tokens
- Role-based permission enforcement at the API level (see §8)
- Password reset, account lockout, session/token expiry

- Employee Management — P0

Single source of truth for the employee record, across the full lifecycle.

- Personal details, employment history, education, next of kin, emergency contact
- Certifications, documents, skills, promotion history
- Department/role/manager assignment, employment type, salary grade, status (active/suspended/exited)

- Department & Org Structure— P0

Model the org chart that everything else (approvals, attendance, payroll) hangs off of.

- Department CRUD with manager assignment and budget field
- Employee-to-department mapping; supports MDA-style or branch-style hierarchies

- Attendance (Core) — P0

Replace manual sign-in sheets with verifiable digital clock-in/out.

- Clock in/out via QR code and GPS geofence verification
- Late-arrival calculation, overtime capture, shift management, holiday schedule, remote-work tracking

- Leave Management — P0

Digitize the full leave lifecycle: request → approval → balance.

- Leave types: annual, casual, study, compassionate, maternity, paternity, medical
- Multi-level approval workflow, running leave balance, calendar visibility for managers

- Payroll (Basics) — P0

Automate the monthly pay run.

- Auto-calculates basic salary, housing allowance, transport allowance, tax (PAYE), pension, NHF, bonuses, overtime, loan deductions, leave deductions, net pay
- Generates PDF payslips and emails them; produces payroll summary reports
- Statutory rates (tax bands, pension %, NHF %) must be admin-configurable, not hardcoded — see §13.

- Dashboards & Reporting (Basic) — P0

Give every role a home screen with the numbers relevant to them.

- Role-specific dashboards (Employee / HR / Admin)
- Exportable reports: payroll, attendance, leave

7.2 Phase 2 — Enterprise Features

- Recruitment (Core) — P1

Move hiring from email into a trackable pipeline.

- Job posting, public application form, resume/CV upload
- Applicant tracking through pipeline stages, interview scheduling, offer-letter generation
- AI-based CV scoring is a separate Phase 3 capability layered on top — see §7.3.

- Performance Management** — P1

Formalize goal-setting and review cycles.

- Quarterly review cycle, goal/KPI setting, manager feedback, 360° review support
- Performance trend view; manual promotion-recommendation notes
- *Feeds the Phase 3 Promotion Recommendation Engine.*

- Learning Portal** — P1

Self-service skill-building, tied back to the employee record.

- Course catalog, assessments, certificate issuance, learning-hours tracking

Document Management — P1

Centralize employment paperwork.

- Offer letters, contracts, promotion letters, warning letters, certificates, signed documents
- Versioning and access control by document sensitivity

Notifications — P1

Keep people informed without them checking manually.

- In-app and email notifications; SMS for time-critical items (leave decisions, interview invites, payslip-ready)

Asset Management— P1

Track company property issued to staff.

- Asset catalog (laptop, ID card, vehicle, phone, keys, uniform), assignment/return workflow, maintenance history
- Linked to the exit-clearance checklist

Digital Employee ID— P1

Verifiable digital identity for every employee.

- QR-coded ID, scan-to-verify; links to attendance check-in and asset issuance

7.3 Phase 3 — Innovation Layer

AI HR Assistant — P1, flagship

Instant answers instead of a ticket to HR.

- Natural-language Q&A over an employee's own data ("How many leave days do I have?")
- Manager-scoped analytical queries ("Who has the highest absenteeism this month?"), respecting RBAC
- Embedded conversational interface in the portal

Predictive Attrition ("Resignation Risk") — P2, showcase

Early warning signal for flight risk, not an automated decision.

- ML score from attendance, performance, tenure, and compensation signals; contributing factors; retention suggestions
- Kept explicitly advisory — see the cold-start and trust notes in §13.*

Smart Recruitment (CV–JD Matching) — P1

Cut screening time with a ranked shortlist.

- Parses uploaded CVs, scores compatibility against the job description, ranks applicants

Employee Sentiment Analysis — P2

Early-warning system for morale, without exposing individuals.

- Anonymous pulse surveys, NLP-based sentiment/happiness scoring, department-level burnout flagging
- Anonymity design (minimum group size before a score is shown) is a hard requirement .

Promotion Recommendation Engine — P2

A data-backed shortlist ahead of promotion cycles.

- Scores attendance, performance, training, experience, certifications, and goal achievement
- Produces a ranked list for HR review — advisory, not automatic

Smart Attendance (Enhancement) — P1/P2

Raise attendance integrity and support field/remote staff.

- Face-recognition clock-in (P1 fast-follow to core attendance)
- Offline capture with sync-on-reconnect (P2 — conflict handling is non-trivial, see §13)
- *Face recognition is always an addition to QR/GPS, never a replacement — see §9 on accessibility and failure fallback.*

Executive Dashboard — P1

One screen for a Director/CEO to see the workforce as a business.

- Headcount, gender ratio, attendance rate, payroll cost & trends, top performers, hiring status & trends, attrition/turnover, promotion trends, training completion, department productivity, leave statistics, revenue-per-employee (private sector)

Automated Workflows — P1

Remove manual hand-offs from multi-step processes.

- Configurable trigger → action chains

- Reference workflow: new hire → create account → assign department → assign laptop → generate digital ID → send welcome email → enroll in onboarding training → notify manager, fully automatic

Multi-Organization Support (Multi-Tenancy) — architectural decision in Phase 1; user-facing in Phase 3

One deployment serving multiple isolated organizations (separate MDAs, or consulting-firm clients).

- Tenant-isolated data, tenant-scoped admin, shared codebase

7.4 Representative user stories

- As an Employee, I want to check my leave balance and request time off from my phone, so I don't have to fill a paper form and wait on HR.
- As a Manager, I want to see who's clocked in and who's on leave today, so I can plan my team's workload.
- As an HR Officer, I want applicants ranked by fit against the job description, so I can shortlist faster.
- As a Payroll Officer, I want tax, pension, and NHF calculated automatically, so payslips are accurate without a manual spreadsheet reconciliation.
- As a Director, I want one dashboard showing headcount, payroll cost, and attrition, so I can make staffing decisions without waiting on a report.
- As a Super Admin, I want to onboard a new MDA or company as an isolated tenant, so its data never mixes with another organization's.
- As an Applicant, I want to apply online and track my application status, so I'm not calling HR to ask what's happening.
- As an HR Officer, I want departments with rising attrition risk flagged early, so I can act before losing good people.

8. Roles & Permissions

Role	Employee Records	Attendance & Leave	Payroll	Recruitment	Performance, Training & Documents	Analytics & Reports	Platform Admin
Super Admin	Full	Full	Full	Full	Full	Full	Full
HR Officer	Full (org-wide)	Manage / approve	View only	Full	Full	HR-scoped	—
Manager	View own team	Approve own team	—	View own postings	Manage own team	Team-scoped	—
Payroll/Finance Officer	View relevant fields	View (for pay calc)	Full	—	Payslips/contracts only	Payroll-scoped	—
Employee	View/edit own	Own only	View own payslips	Own applications	Own only	—	—

| Director/Executive | View-only, org-wide | View aggregate | View aggregate cost | View aggregate | View aggregate | Full (Executive Dashboard) | — |
| Applicant | — | — | — | Own application only | — | — | — |

Permissions must be enforced server-side on every API endpoint, not only hidden in the frontend UI — a manager's JWT should be structurally incapable of pulling another department's payroll data, not just prevented by the interface from requesting it.

9. Non-Functional Requirements

Performance

- Page loads under 2 seconds on typical broadband/4G; dashboard queries under 3 seconds even against multi-year attendance/payroll history
- Long-running work (payroll runs, report generation, AI scoring) goes through the background job queue (Celery) so it never blocks the API

Scalability

- Must scale from a single pilot department to a full state civil service (thousands of employees) without re-architecture
- Multi-tenant from the data layer up (see §7.3)

Security

- TLS in transit; encryption at rest for PII and payroll data
- RBAC enforced at the API layer (see §8)
- Audit logging for sensitive actions: salary changes, record deletions, permission changes
- JWT refresh-token rotation; account lockout after repeated failed logins

Data Privacy & Compliance (Nigeria)

- WorkForge processes personal data under the Nigeria Data Protection Act (NDPA) 2023, enforced by the Nigeria Data Protection Commission (NDPC)
- Public-sector deployment likely falls into a category the NDPC treats as higher-risk — confirm data-controller/processor registration obligations before pilot go-live
- Biometric data (face recognition) is sensitive/special-category data: requires explicit informed consent and a documented lawful basis before that feature is switched on. Face recognition should never be the *only* clock-in method — QR/GPS remain the fallback, which also covers accessibility and camera/lighting failure cases
- Confirm data residency/hosting expectations with government stakeholders before selecting a cloud region
- Employees must be able to see what data WorkForge holds about them and request correction, from the Employee Portal itself
- Keep AI-driven people-decisions (attrition risk, promotion recommendations) advisory and human-reviewed, not automated actions

Availability & Reliability

- Target 99%+ uptime during pilot, 99.5%+ once payroll-critical
- Payroll runs must be idempotent and resumable — a retry should never double-pay or silently drop an employee

Usability & Accessibility

- Mobile-responsive for field and remote workers
- Usable on low-bandwidth connections; attendance supports offline capture with later sync
- Reasonable contrast/keyboard-navigation support, appropriate for public-sector accessibility expectations

Auditability

- Every payroll change, leave approval, and permission change is attributable to a user, timestamped, and immutable in the audit log

10. System Overview & Technology Stack

Full component diagrams, database schema, folder structure, and API route list live in the companion architecture document (`WorkForge\_Architecture.docx`) — that remains the source of truth for implementation design. This section is a summary for context.

Architecture, in one line: React frontend (role-based portals) → Django REST API (service-oriented modules: auth, employee, leave, payroll, recruitment, attendance, notifications, reports, analytics, AI assistant) → PostgreSQL, with Redis, Celery, cloud file storage, and email/SMS/PDF/face-recognition/QR services supporting it.

Layer	Technology
Frontend	React + TypeScript
Styling	Tailwind CSS
State management	Redux Toolkit + React Query
Backend	Django + Django REST Framework
Authentication	JWT
Database	PostgreSQL
Cache	Redis
Background jobs	Celery
File storage	AWS S3 or Cloudinary
Search	PostgreSQL full-text search (Elasticsearch optional)
Charts	Recharts
Reports	ReportLab
Containerization	Docker
CI/CD	GitHub Actions
Deployment	Nginx + Unicorn

11. Data Model Summary

Full schema detail lives in the architecture doc; this is the entity list for reference.

Entity	Purpose
Employee	Core person record — identity, employment, status
Department	Org unit — manager, budget, hierarchy
Role	Job role/title, tied to salary grade
Attendance	Daily clock-in/out records
LeaveRequest	Leave applications and approval state
Payroll / Salary / Benefits	Pay run records, salary structure, deductions
Recruitment / JobApplication	Job postings and applicant pipeline
PerformanceReview	Review cycles, goals, ratings
Training	Courses, assessments, certificates
Asset	Company property assignment and history
Documents	Contracts, letters, certificates
Notification	In-app/email/SMS notification records
AuditLog	Immutable record of sensitive actions
ExitInterview	Offboarding record

12. Release Plan & Milestones

Phase	Exit criteria (what "done" looks like)	Suggested pacing*
Phase 1	A new hire can be onboarded, clock in/out via QR + GPS, request and get leave approved, and receive an accurate auto-calculated payslip — end to end, for at least one pilot department	~8–10 weeks
Phase 2	A candidate can apply, be interviewed, get hired, and flow automatically into the Phase 1 employee record; assets, documents, and notifications all work against real records	~6–8 weeks
Phase 3	The AI Assistant correctly answers real leave/attendance questions; the Executive Dashboard reflects live cross-module data; at least one predictive feature is demoable on realistic pilot data	~6–8 weeks

13. Risks, Assumptions & Open Questions

Risks

Scope. 20+ modules and features across three phases is a large build for a typical capstone timeline — prioritize ruthlessly within each phase rather than building every listed capability at equal depth.

Cold start for AI/ML features. Predictive attrition and promotion recommendations need historical data to be meaningfully accurate; a new pilot won't have years of attrition history. Consider launching these as rule-based/heuristic scoring first, and be transparent that model quality improves as real data accumulates.

Biometric accuracy & compliance. Face-recognition accuracy varies with lighting and camera quality, and carries the data-protection obligations noted in §9 — always keep a non-biometric fallback.

Offline sync. True offline-first attendance with later sync introduces conflict-resolution complexity (duplicate check-ins, clock drift) that's easy to underestimate.

-Change management. Moving government MDA staff from paper to digital processes is a change-management effort as much as a software rollout — pilot with one MDA or department before a wider push.

Statutory rates. PAYE tax bands, pension contribution rates, and NHF rates are set by regulation and change periodically; hardcoding them will eventually break payroll.

Compliance timing. Biometric data plus government employee data likely brings WorkForge into a higher-scrutiny NDPC category — get a compliance review done before pilot go-live, not after.

Assumptions

- Single-country (Nigeria) statutory payroll rules for v1.
- The pilot starts with one MDA/department or one company, not a state-wide rollout on day one.
- Users have at least basic smartphone or desktop access, with offline support for attendance in low-connectivity settings.

Open questions

- Which specific MDA (or private organization) is the intended Phase 1 pilot partner, if any — and is there existing payroll/employee data to migrate?
- What are the current PAYE tax bands, pension contribution rate, and NHF rate to encode as configurable defaults?
- Is there a target submission/defense date that should anchor the phase timeline in §12?
- Should the AI HR Assistant run on a hosted LLM API or a self-hosted model, given data sensitivity and budget?
- What's the minimum pilot-department size needed for a believable demo of the analytics/AI features?

14. Appendix: Glossary & Reference Documents

Glossary

-MDA— Ministries, Departments and Agencies (Nigerian government structure)

- NDPA — Nigeria Data Protection Act, 2023
- NDPC — Nigeria Data Protection Commission (NDPA's enforcement body)
- NHF — National Housing Fund
- PAYE — Pay As You Earn (income tax deducted at source)
- RBAC — Role-Based Access Control
- JWT — JSON Web Token
- KPI — Key Performance Indicator